Shantnu Kaushal

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EDUCATION

St. John's University Queens, NY
Bachelor of Science, Computer Science *Aug 2022 – Jan 2026*

- **GPA**: 4.0
- **Relevant Coursework:** Software Engineering, Analysis of Algorithms, Database Management Systems

PROFESSIONAL EXPERIENCE

Sports Excitement *Apr 2025 – Present*
Software Engineer

- Lead delivery of 15+ core features in an Agile/Scrum environment, partnering with stakeholders to translate user requirements into a production-ready social networking platform
- Architect a high-performance backend using Node.js/Express and PostgreSQL, optimizing schema design and indexing to reduce query latency by 25% for complex datasets
- Drive platform deployment on Linux, streamlining CI/CD pipelines to improve overall reliability and reduce production turnaround time by 40%

NASA L'Space *Sept 2025 – Dec 2025*
Student Researcher

- Engineered a theoretical signal processing framework to mitigate Radio Frequency Interference for lunar missions, architecting an adaptive filtering algorithm targeting a 20 dB suppression threshold
- Designed a hybrid real-time architecture utilizing C++ for high-speed noise cancellation and Python for machine learning integration to recover faint astronomical signals in high-noise environments
- Authored a New Technology Report for a competitive NASA solicitation, defining the software integration strategy and technology infusion pathways for future lunar missions

PROJECTS

Neura | *TensorFlow, YOLOv8, OpenCV, FastAPI, Docker*

- Built a computer vision application that identifies 8 deep-sea organism classes from uploaded underwater imagery through a YOLOv8 detection workflow
- Refined a TensorFlow detector on 1,057 labeled training images with class balancing and image augmentation for underwater object detection
- Integrated model inference into FastAPI to return organism classifications, confidence scores, and bounding-box coordinates

TweetCheck | *PyTorch, BERT, Apache Kafka, Go, Redis*

- Fine-tuned a BERT model with PyTorch to classify tweet sentiment across incoming social media posts with low-latency inference
- Constructed a high-throughput streaming pipeline that processes 1.6M+ social media posts through Go and Kafka at 1,000+ events/sec
- Cached model outputs, event counts, and processing lag in Redis for fast dashboard reads and real-time pipeline visibility

ReSource | *Python, all-MiniLM-L6-v2, pgvector, Flask, Next.js*

- Ingested PDFs by extracting text, chunking source material, and generating all-MiniLM-L6-v2 embeddings for document-grounded retrieval workflows
- Matched user questions to the closest PDF chunks by running pgvector cosine similarity search across stored document embeddings
- Generated grounded answers from selected PDF context through a Flask API, keeping responses tied to retrieved source material

TECHNICAL SKILLS

Programming & Frameworks: Python, Go, C++, Java, SQL, TypeScript/JavaScript, Node.js, Express, FastAPI, Next.js, React
AI & ML: PyTorch, TensorFlow, Transformers, Computer Vision, RAG, NLP, Vector Search, Pandas, NumPy, Scikit-learn
Infrastructure & Tooling: Apache Kafka, Docker, PostgreSQL, pgvector, Redis, Supabase, MinIO, Linux, Prisma